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Professional Summary Solutions-driven AI/ML Developer and Software Engineer with expertise in Python, Data Engineering, and full-stack development with proven track record of processing 24GB+ datasets and architecting complex web applications and with experience at building AI-powered tools and optimizing algorithms to solve real-world operational challenges

Technical Skills

- **Programming Languages:** Python, Java, SQL, C#, JavaScript
- **Cloud Concepts & DevOps:** AWS (Theoretical Knowledge), Git, GitHub
- **Data Architecture:** Data Cleaning, Wrangling Techniques, Predictive Modeling
- **AI & ML:** Prompt Engineering, Generative AI Integration, AI Bot Development

Experience AI/Data Science Intern | 1M1B Green Internship — Virtual, TN | December 2025 – January 2026

- **Engineered a scalable $O(n)$ data pipeline** to process 24GB+ of environmental time-series data, reducing computational complexity by 300x through optimized vectorized operations
- **Architected an interactive web application** deployed for real-time water quality monitoring across 49 Asian countries, aligning risk assessments with WHO/EPA standards
- **Deployed predictive machine learning models** (CatBoost/XGBoost) utilizing k-fold cross-validation to ensure stability across 7 key environmental indicators
- **Designed dynamic visualization dashboards** integrating live API data from the Hong Kong EPD into a responsive database, implementing advanced data processing algorithms

Projects

Roots – AI-Powered Code Editor & Portfolio Platform |

- Secured 2nd place out of 20 teams in the Sustainability category at the Pragyotsav Project Expo (Easwari Engineering College) for developing an open-source web IDE
- Engineered a Smart AI Router to dynamically select optimal foundational LLMs (Gemini, Claude, DeepSeek) based on task complexity for real-time code generation

- Integrated multi-language execution services (Python, Java, JavaScript) utilizing the Monaco Editor to mirror advanced VS Code functionality within the browser
- Implemented secure OAuth 2.0 flows for GitHub, LinkedIn, and Twitter, enabling seamless direct repository commits and automated portfolio updates
- Optimized frontend performance with Vite and utilized Zustand for global state management, ensuring highly responsive interactions across the Learning Hub interface

Certifications

- **NPTEL (IIT):** Natural Language Processing (NLP), Introduction to Internet of Things (IoT), Blockchain Architecture
- **IBM:** Make Agentic AI Work for you, Fundamentals of Sustainability and Technology, ML & Deep Learning
- **Cisco:** Security & Connectivity, Python Essentials, CyberSecurity, Modern AI
- **Microsoft:** Foundational C# Certification

Languages

- **English:** Native / Bilingual Proficiency
- **Tamil:** Native / Bilingual Proficiency
- **Hindi:** Elementary Proficiency(Comprehension)

Technical Interests

- Algorithm Optimization (LeetCode) and exploring new algorithms
- Hardware Benchmarking (PC components, emerging smartphone tech)
- AI Applications and seamless LLM integration strategies